ECE 4100
Project 3 Out of Order Superscalar Processor
Experiments Report
Yue Pan
04/17/2021

Part 1

Determine the appropriate number of function units and the fetch rate for each benchmark for the given number of ROB entries and PREGS. This means that the J, K, L and F parameters should be varied and minimized for default values of R and P and I.

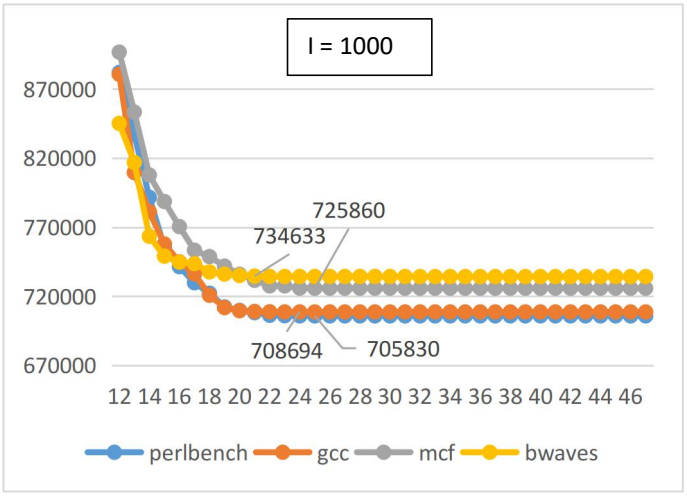
Before performing the experiments, we should first determine the ranges of parameters. For fetch rate, it won't make sense for F to exceed the ROB size 12, which is also the effective size of scheduling queue. For the max number of function units, they should not exceed the maximum possible usage of function units. For add units, it's max value should be 12, for multiply, it should be 4, and for LW units, 6.

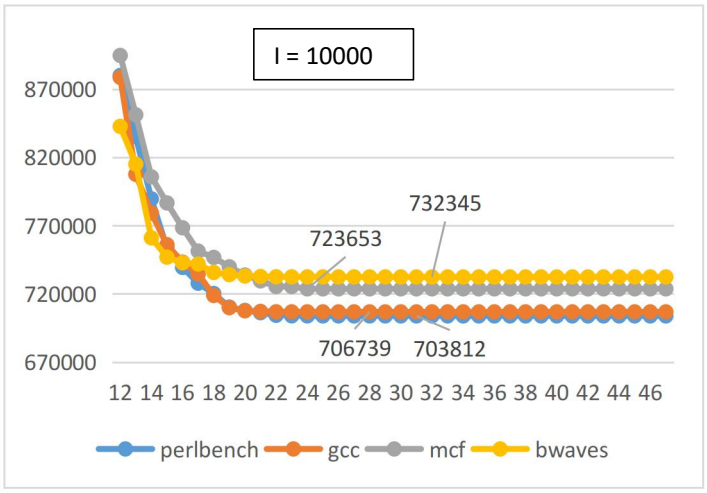
Part 2

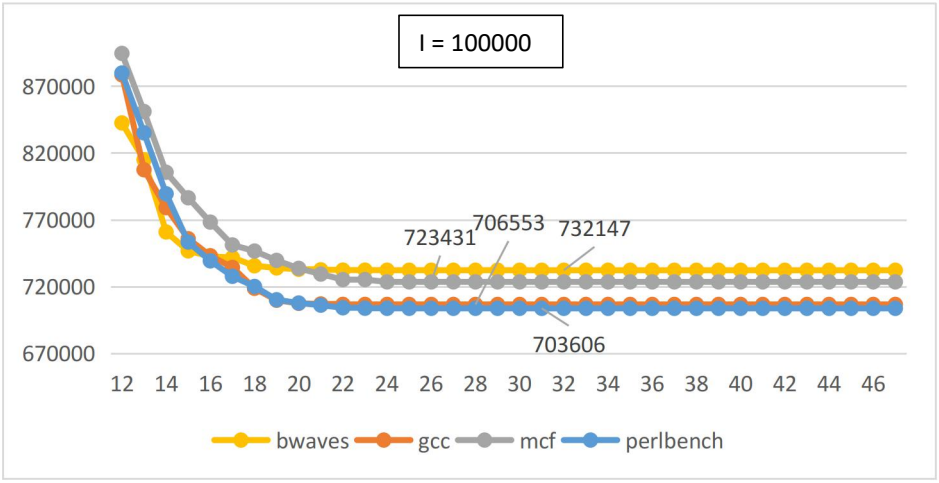
For the given default J, K, L, F, and P parameters find the optimum ROB size when interrupts occur at every 1000, 10000, 100000 cycles.

Before experiments, I expect that to see the trend that the maximum cycles count goes down with increasing ROB size but stops decreasing at some point. This is because that a bigger ROB size help better utilizing the function units and improve instruction level parallelism. However, when ROB gets too big, the performance will be, in turn, limited by the number of function units. At this point, a bigger ROB does not bring any performance improvements.

This part is relatively straightforward. We will just start increasing the size of ROB from 12. The resulting graphs and optimal ROB sizes are shown below.







Optimal ROB sizes for I values and traces:

Interrupt Frequency	bwaves	gcc	mcf	perlbench
I = 1000	32	24	25	25
I = 10000	32	28	24	31
I = 100000	32	28	26	31